



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Course Name: General physics B (II) / A Dept.: PHYSICS

Exam Duration: 2 hours Exam Paper Setter: Physics teaching team

Question No.	1	2	3	4	5	6	7	8	9	10
Score	30	6	10	10	12	10	12	10		

This exam paper contains 8 questions and the score is 100 in total. (Please hand in your exam paper, answer sheet, and your scrap paper to the proctor when the exam ends.)

Q1. Multiple Choice Questions (3 points each), there is only one correct answer for each question.

1) A $2.0 \mu\text{F}$ and a $1.0 \mu\text{F}$ capacitor are connected in series and a potential difference is applied across the combination. The $2.0 \mu\text{F}$ capacitor has:

- A) twice the charge of the $1.0 \mu\text{F}$ capacitor
B) half the charge of the $1.0 \mu\text{F}$ capacitor
C) twice the potential difference of the $1.0 \mu\text{F}$ capacitor
D) half the potential difference of the $1.0 \mu\text{F}$ capacitor

$$C = \frac{Q}{U}$$

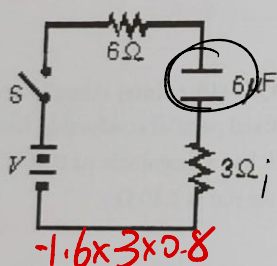
$$U = \frac{Q}{C}$$

2) In the circuit shown, the capacitor is initially uncharged, and $V = 10 \text{ V}$. At time $t = 0$, switch S is closed.

If τ denotes (代表) the time constant, the current through the 3Ω resistor when $t = \tau/10$ is:

- A) 0.50 A
B) 0.75 A
C) 1.0 A
D) 1.5 A

$$\frac{10}{9} e^{-\frac{1}{10}}$$



$$i = \frac{\varepsilon}{R} e^{-\frac{t}{\tau}}$$

3) At one instant an electron (charge $= -1.6 \times 10^{-19} \text{ C}$) is moving in the xy plane, the components of its velocity being $v_x = 5.0 \times 10^5 \text{ m/s}$ and $v_y = 3.0 \times 10^5 \text{ m/s}$. A magnetic field of 0.80 T is in the positive x direction. At that instant the magnitude of the magnetic force on the electron is:

- A) 0 N B) $3.8 \times 10^{-14} \text{ N}$ C) $6.0 \times 10^{-14} \text{ N}$ D) $6.4 \times 10^{-14} \text{ N}$

$$v = \langle 5, 3 \rangle \times 10^5$$

$$F = qvB \sin \theta$$

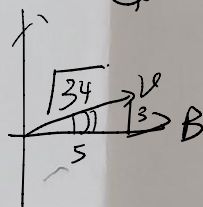
$$= e v B$$

4) The Hall effect can be used to calculate the charge-carrier number density in a conductor. If a conductor carrying a current of 2.0 A is 0.5 mm thick, and the Hall effect voltage is $4.5 \mu\text{V}$ when it is in a uniform magnetic field of 1.2 T , what is the density of charge carriers in the conductor?

- A) $1.0 \times 10^{28} / \text{m}^3$ B) $6.7 \times 10^{27} / \text{m}^3$ C) $4.6 \times 10^{27} / \text{m}^3$ D) $1.7 \times 10^{27} / \text{m}^3$

$$n = \frac{Bi}{VLe}$$

$$\frac{1.2 \times 2}{4.5 \times 0.5 \times 10^{-3}}$$



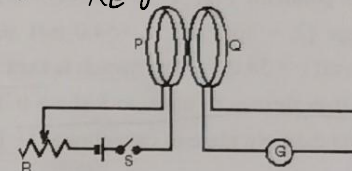
$$\frac{\mu_0 I_1 I_2}{2\pi d}$$

5) Two parallel wires, 4.0 cm apart, carry currents of 2.0 A and 4.0 A respectively, in opposite directions. The force per unit length of one wire on the other is:

- A) $1 \times 10^{-3} \text{ N/m}$, repulsive B) $1 \times 10^{-3} \text{ N/m}$, attractive
C) $4 \times 10^{-5} \text{ N/m}$, repulsive D) $4 \times 10^{-5} \text{ N/m}$, attractive

6) Coils P and Q each have a large number of turns of insulated wire. When switch S is closed, the pointer of galvanometer G is deflected (偏转) toward the left. Now that S is closed, to make the pointer of G deflect toward the right, we could:

- A) move the slide of the rheostat (变阻器) R quickly to the right
B) move coil P toward coil Q
C) move coil Q toward coil P
D) open S



7) A single loop of wire with a radius of 7.5 cm rotates about a diameter in a uniform magnetic field of 1.6 T . To produce a maximum emf of 1.0 V , it should rotate at:

- A) 0 rad/s B) 2.7 rad/s C) 5.6 rad/s D) 35 rad/s



$$\mathcal{E} = B \omega \pi R^2$$

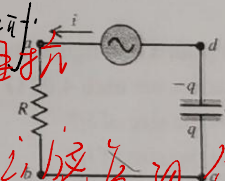
发电机

8) As shown in figure, $R = 200 \Omega$, $C = 15.0 \mu\text{F}$, $f_d = 60.0 \text{ Hz}$, and $\mathcal{E}_m = 36.0 \text{ V}$. What is the current amplitude I ?

- A) 0.135 A
B) 0.231 A
C) 0.357 A
D) 0.625 A

$$I = \frac{\mathcal{E}_m}{\sqrt{R^2 + (\omega C)^2}}$$

$$\omega = \frac{2\pi}{T} = 2\pi f$$



9) The magnetic field in a sinusoidal light wave has an amplitude of $3.3 \times 10^{-7} \text{ T}$. The intensity of the wave is:

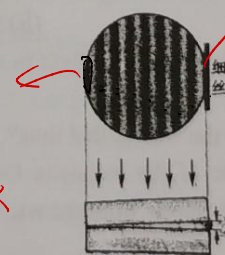
- A) $1.7 \times 10^{-4} \text{ W/m}^2$ B) 13 W/m^2 C) 27 W/m^2 D) $1.0 \times 10^5 \text{ W/m}^2$

$$I = \frac{B_m^2}{2\mu_0}$$

$$E_B = \frac{B_m}{\sqrt{\epsilon_0 \mu_0}}$$

10) In figure, Clip a D diameter filament to one side of two flat glass bricks, under the vertical irradiation (照射) of yellow light (590 nm), interference fringes are formed as shown. What is diameter D?

- A) $2.36 \mu\text{m}$
B) $2.98 \mu\text{m}$
C) $3.26 \mu\text{m}$
D) $3.58 \mu\text{m}$



$$2D = 8\lambda$$

$$\frac{9}{2} \lambda$$

弄清 m 的级数
m 级

Q2

1.



$$mg = \frac{IA}{C} \quad \rho \frac{4}{3} \pi R^3 g = \frac{I \pi R^2}{C}$$

$$\rho v g = \frac{IA}{C} \quad \left(\rho \frac{4}{3} R c g \right) = I = \left[\frac{P}{4 \pi d^2} \right]$$

$$P = \rho \frac{4}{3} R c g \cdot 4 \pi d^2 = \rho \frac{16 \pi}{3} R c g d^2$$

$$= 19 \times \frac{10^{-3}}{10^{-6}} = 19 \times 10^3 \times \frac{16}{3} \pi \times 2 \times 10^{-3} \times 3 \times 10^8 \times 9.8 \times 10^{-4}$$

$= 4.18 \times 10^6 \text{ N}$

$$x = Q_m \cos(\omega t + \varphi)$$

$$\dot{x} = -Q_m \omega \sin(\omega t + \varphi)$$

当 $t=0$ 时

$$\frac{\dot{x}_0}{Q_0} = \omega \tan \varphi \quad \text{故 } \tan \varphi = \frac{\dot{x}_0}{\omega Q_0} \quad \text{故 } \varphi = 0.7853 \text{ rad (7)}$$

$$x = Q_m \cdot \omega \sin \varphi \quad \text{故 } Q_m = 0.051756 \text{ (E)}$$

$$Q_{t_0} = Q_m \cos(\omega \times 0.3 + \varphi) = -0.00505 \text{ c}$$

$$\text{故 } Q_{t_0} = 0.00505 \text{ c}$$

Q3. (10 points) An infinite nonconducting sheet has a surface charge density $\sigma = +5.80 \text{ C/m}^2$. (a) How much work is done by the electric field due to the sheet if a particle of charge $q = +1.60 \times 10^{-19} \text{ C}$ is moved from the sheet to a point P at distance $d = 3.56 \text{ cm}$ from the sheet? (b) If the electric potential V is defined to be zero on the sheet, what is V at P ?

Q4. (10 points) In figure, the ideal batteries have emfs $\mathcal{E}_1 = 12.0 \text{ V}$ and $\mathcal{E}_2 = 4.00 \text{ V}$, and the resistances are each 4.00Ω .

(a).



$$E = \frac{\sigma}{2\epsilon_0} \quad W = qEd = \frac{q\sigma}{2\epsilon_0} \times d$$

代数 $W = 1.87 \times 10^{-2} \text{ J}$

(b)

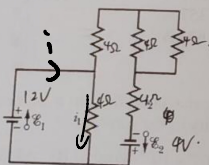
$$V = -Ed \quad \text{代数 } V = -1.13 \times 10^{-2} \text{ V}$$

Q4. (10 points) In figure, the ideal batteries have emfs $\mathcal{E}_1 = 12.0 \text{ V}$ and $\mathcal{E}_2 = 4.00 \text{ V}$, and the resistances are each 4.00Ω .

(a) What is the size of i_1 ?

(b) What is the size of i_2 ?

(c) What is the energy transfer rate of battery 1?

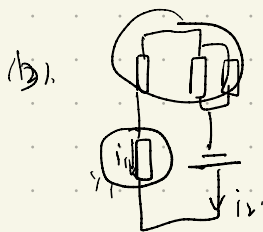


$$\sin \phi = \frac{P}{P_{\text{max}}}$$

从负到正: $+\mathcal{E}$

(a) $i_1 R = \mathcal{E}_1$

故 $i_1 = \frac{12.0 \text{ V}}{4.00 \Omega} = 3.00 \text{ A}$



$$R_B = 2R + \frac{RR}{R+R} = 10\Omega$$

$$\mathcal{E}_2 + i_1 R - i_2 R_B = 0$$

故 $i_2 = 1.60 \text{ A}$

(c).

$$P = \mathcal{E}_1(i_1 + i_2)$$

故 $P = 12.0 \text{ V} \times 4.6 \text{ A} = 55.2 \text{ W}$

Q5. (12 points) A silver wire has resistivity $\rho = 1.62 \times 10^{-8} \Omega \cdot \text{m}$ and a cross-sectional area of 6.00 mm^2 . The current in the wire is uniform and changing at the rate of 1500 A/s .

(a) What is the magnitude of the (uniform) electric field in the wire when the current in the wire is 80 A ?

(b) What is the displacement current in the wire at that time?

(c) What is the ratio of the magnitude of the magnetic field due to the displacement current to that due to the current at a distance r from the wire? Assume the current is also 80 A .

(a) $E = \rho J = \rho \frac{I}{A}$

(b).

$$\phi_E = \rho I$$

代数: $E = 0.216 \text{ V/C}$

$$i_d = \epsilon_0 \frac{d\phi_E}{dt} = \epsilon_0 \frac{dI}{dt}$$

故 $i_d = 2.15 \times 10^{-6} \text{ A}$

(c).

$$\text{ratio} = \frac{B_d}{B_r} = \frac{B_d 2\pi r}{B_r 2\pi r} = \frac{i_d}{i}$$

代数: $\text{ratio} = 2.69 \times 10^{-8}$

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Q6. (10 points) Light of wavelength 600 nm is incident normally on a double slits. The separation between adjacent slits is $6.00\mu\text{m}$, the slit width is $2.00\mu\text{m}$, the viewing screen is at distance $D = 4.00\text{ m}$. Let a y axis extend upward along the viewing screen, with its origin at the center of the pattern, let I_p present the intensity at point P at $y = 12.0\text{ cm}$.

(a) What the ratio of I_p to the intensity I_m at the center of the pattern?

(b) What is the second largest value of the order number m of the maxima produced by the slits? explain the reasons.

Slit Diffraction Intensity:

$$I(\theta) = I_m (\cos^2 \beta) \left(\frac{\sin \alpha}{\alpha} \right)^2$$

$$\text{where } \alpha = \frac{\pi a}{\lambda} \sin \theta \text{ and } \beta = \frac{\pi d}{\lambda} \sin \theta$$

(a) $d = 6\mu\text{m}$ $a = 2\mu\text{m}$

$$\sin \theta \approx \tan \theta = \frac{y}{D}$$

$$\alpha = \frac{\pi a y}{\lambda D} \quad \beta = \frac{\pi d y}{\lambda D}$$

代入得: $\alpha = 0.314\text{ rad}$ $\beta = 0.942\text{ rad}$

$$\frac{I_p}{I_m} = \cos^2 \beta \left(\frac{\sin \alpha}{\alpha} \right)^2$$

代入得: $\frac{I_p}{I_m} = 0.334$

(b)

$$d \sin \theta = m_1 \lambda$$

$$a \sin \theta = m_2 \lambda$$

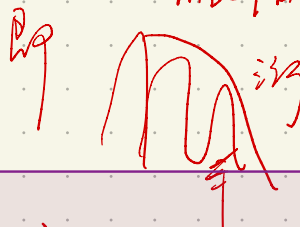
注意: 干涉 $m = 0, 1, 2, \dots$ 衍射 $m = 1, 2, \dots$ 暗

$$d \sin \theta = m_1 \lambda$$

$$a \sin \theta = m_2 \lambda$$

θ 与 y (坐标) 一一对应

因此 $m_2 = 1$ 时, $m_1 = \frac{d}{a} > 1$



(b)

$$d = m \lambda$$

$$m = \frac{d}{\lambda} = 10$$

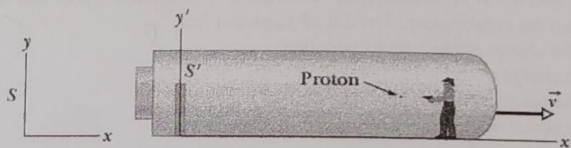
由于 $m=10$ 在无穷远取到

$m=9$ 是干涉暗纹

故次大为 $m=7$

Q7. (12 points) Figure shows a ship (attached to reference frame S') passing us (standing in reference frame S) with velocity $\vec{v} = 0.950c\hat{i}$. A proton is fired at speed $0.980c$ relative to the ship from the front of the ship to the rear. The proper length of the ship is 760 m.

- (a) What is the temporal (时间的) separation between the time the proton is fired and the time it hits the rear wall of the ship according to a passenger in the ship?
- (b) What is that temporal separation according to the proton?
- (c) What is that temporal separation according to us?



(a) $t' = \frac{L'}{v_p}$ $\Delta t' = t' = 2.58 \times 10^{-6}\text{ s}$

(b) $L_p = \sqrt{1 - \frac{v_p^2}{c^2}} L'$ $t_p = \frac{L_p}{v_p} = 5.14 \times 10^{-7}\text{ s}$ 或 $t' = \frac{t_p}{1 - \frac{v_p^2}{c^2}}$

(c) $L = \sqrt{1 - \frac{v^2}{c^2}} L'$

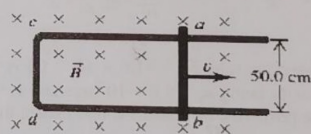
$$v_1 = v = 0.95c$$

$$v_2 = \frac{-v_p + v}{1 - \frac{v_p v}{c^2}} = \frac{-0.03c}{1 - 0.95 \times 0.98} = -0.435c$$

$$u = \frac{u' + v}{1 + u'v/c^2}$$

$$t = \frac{L}{|v_1| + |v_2|} = 5.71 \times 10^{-7}\text{ s}$$

Q8. (10 points) Assume a conducting rod of length $L = 50.0$ cm is sliding along two fixed parallel conducting frictionless rails which have zero resistances. The uniform field B has a magnitude of 0.800 T, perpendicular to the plane into the page. The resistance of the rod is 1.50Ω .



(a) Find the external force (magnitude) to keep the rod moving at a constant speed 7.50 m/s to the right.

(b) Assume the rod at a constant speed 7.50 m/s to the right has a mass $m = 500$ g, now we suddenly stop applying the external force to the rod. What will be the velocity of the rod after 5.00 seconds?

(b)

$$F_A = -m \frac{dv}{dt} = BIL$$

$$= B \frac{\mathcal{E}}{R} L = B \frac{BLv}{R} L$$

$$= \frac{B^2 L^2 v}{R} = m \frac{dv}{dt}$$

$$\text{则 } \frac{dv}{v} = -\frac{B^2 L^2}{mR} dt$$

$$\ln v \Big|_{v_0}^v = -\frac{B^2 L^2}{mR} t$$

$$\ln \frac{v}{v_0} = -\frac{B^2 L^2}{mR} t$$

$$\text{则 } v = v_0 e^{-\frac{B^2 L^2}{mR} t}$$

$$v_0 = 7.5 \text{ m/s} \quad \text{at } t = 5.00 \text{ s}$$

$$\text{则 } v = 2.58 \text{ (m/s)}$$

(a)

$$F_A = BIL$$

$$I = \frac{\mathcal{E}}{R}$$

$$\mathcal{E} = BLv$$

$$\text{则 } F = F_A$$

$$\text{则 } F = \frac{B^2 L^2 v}{R}$$

代入

$$F = 0.8 \text{ (N)}$$

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